

Technical Requirements (TRD)

Architecture Pattern: Classic three-tier client-server architecture using the **MERN stack** (MongoDB, Express.js, React.js, Node.js). The frontend communicates exclusively with the backend via HTTP/JSON (Axios), and the backend persists data through the Mongoose ODM.

Three-Tier Architecture Diagram

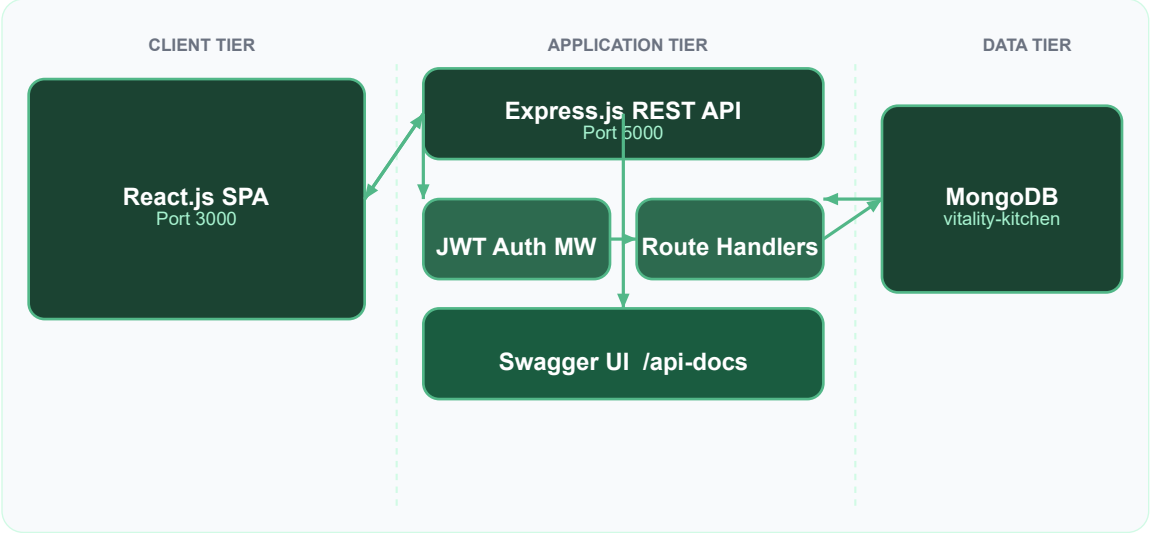


Fig 1.1 — MERN Stack: Client Tier → Application Tier → Data Tier

Tier	Technology	Port	Role
Client Tier	React.js SPA	3000	Browser-rendered Single Page Application
Application Tier	Node.js + Express.js	5000	REST API, JWT middleware, business logic
Data Tier	MongoDB (Mongoose ODM)	27017	NoSQL document persistence
Docs	Swagger UI	5000/api-docs	Interactive OpenAPI 3.0 documentation

Component Descriptions

1. Frontend — React.js SPA

Component	Technology	Purpose
Entry Point	src/index.js	Mounts the React app; wraps in BrowserRouter

Routing	React Router DOM v7	SPA navigation between pages
State Management	React Context API	Global state: AuthContext (JWT) + FavoritesContext
Form Handling	Formik + Yup	Validated forms for login, register, create recipe
HTTP Client	Axios	All API calls from frontend to backend
Styling	Tailwind CSS	Utility-first styling; responsive design
UI Components	MUI (@mui/material)	Supplementary Material UI components
PDF Export	html2pdf.js	Grocery list PDF download from browser

Page Inventory

Page	File	Description
Home / Landing	Home.js	Marketing landing page
Authentication	AuthPage.jsx	Combined login / register form (Formik + Yup)
Recipes	Recipes.js	Browse & filter all recipes with search
Recipe Detail	RecipeDetail.js	Full recipe view + rating widget
Create Recipe	CreateRecipe.js	Authenticated recipe creation form with nutrition preview
Meal Planner	MealPlanner.js	Weekly drag-and-assign planner (Mon–Sun)
Grocery List	GroceryList.js	Shopping list generator + PDF export
Dashboard	Dashboard.js	User statistics overview
Profile	Profile.js	Profile management, favorites list, dietary preferences

2. Backend — Node.js + Express.js

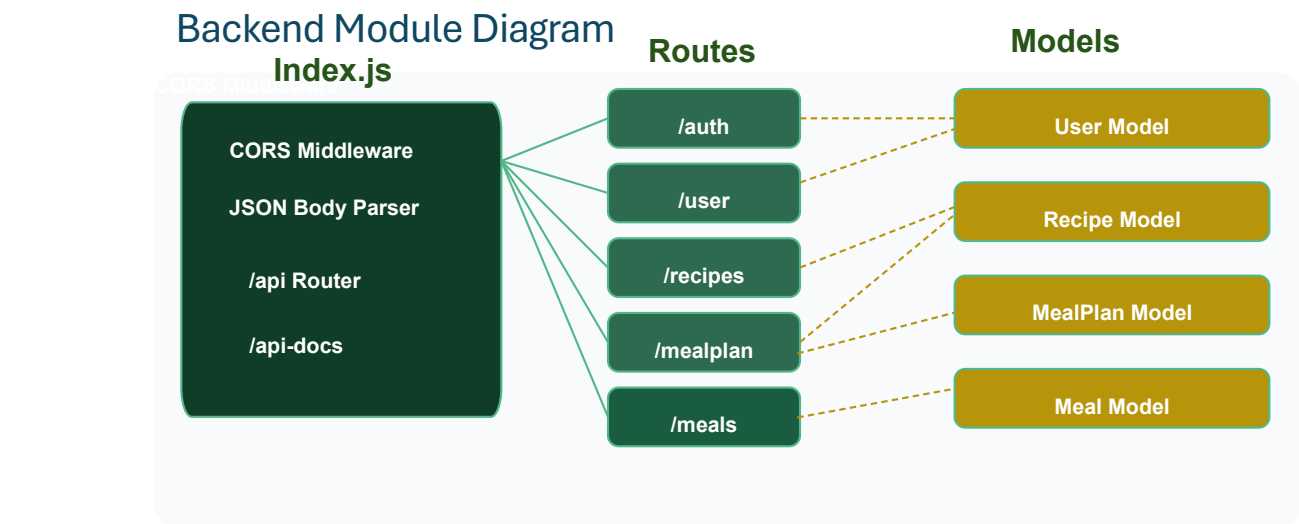


Fig 2.1 — index.js → Route Modules → Mongoose Models

Module	Path	Role
Server entry	src/index.js	Bootstraps Express, connects DB, mounts routes
DB connection	src/db/connection.js	Mongoose connect() using MONGO_URI env var
Auth middleware	src/middleware/auth.js	Validates Authorization: Bearer
Auth routes	src/routes/auth.js	Register + Login handlers
User routes	src/routes/User.js	Profile CRUD, favorites, account deletion
Recipe routes	src/routes/recipe.js	Recipe CRUD, rating, nutrition calculation
Meal plan routes	src/routes/mealPlan.js	Weekly plan CRUD, grocery list generation
Meal routes	src/routes/meal.js	Legacy simple meal entity (backward compat.)
Swagger spec	src/swagger.js	OpenAPI 3.0 definition (18 KB)
DB Seeder	src/db/seedLibraryRecipes.js	Populates recipe library on first run

Infrastructure Requirements

1. Runtime Environment

Requirement	Specification
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Node.js	v18+ (LTS recommended)
npm	v9+
Operating System	Windows / macOS / Linux

2. Database

Requirement	Specification
Database Engine	MongoDB v6+
ODM	Mongoose v9.x
Connection	mongodb://127.0.0.1:27017/vitality-kitchen (local)
Indexes	Compound unique index on MealPlan(user, startDate)
Data Seeding	seedLibraryRecipes.js must be run once on initial setup

3. Network & Ports

Service	Default Port	Configurable Via
Express API server	5000	PORT env variable
React dev server	3000	PORT env variable
MongoDB	27017	MONGO_URI env variable
Swagger UI	5000 (/api-docs)	N/A — same process as API

3.4 Environment Variables (.env — Backend)

Variable	Required	Default	Description
PORT	No	5000	Express server port
MONGO_URI	Yes	—	MongoDB connection string
JWT_SECRET	Yes	—	Secret key for signing JWT tokens

Limit	Value	Reason
JSON body size	15 MB	Supports Base64-encoded recipe images
JWT expiry	7 days	Balances security and user convenience

Development Tools, Frameworks & Libraries

1. Backend Dependencies

Package	Version	Purpose
express	^5.2.1	Web framework — handles routing, middleware, request/response
mongoose	^9.5.0	MongoDB ODM — schema definition, queries, population
jsonwebtoken	^9.0.3	JWT creation and verification
bcrypt	^6.0.0	Password hashing (salt rounds: 10)
cors	^2.8.6	Cross-Origin Resource Sharing middleware
dotenv	^17.4.2	Environment variable loading from .env file
swagger-ui-express	—	Serves interactive Swagger/OpenAPI 3.0 docs at /api-docs

2. Backend Dev Dependencies

Package	Version	Purpose
jest	^30.3.0	Test runner — unit & integration tests with coverage reports
supertest	^7.2.2	HTTP integration testing — makes real requests to Express
nodemon	^3.1.14	Auto-restarts server on file changes during development
eslint	^10.2.1	Code linting — enforces code quality and consistency
prettier	^3.8.3	Code formatting — consistent style across all source files

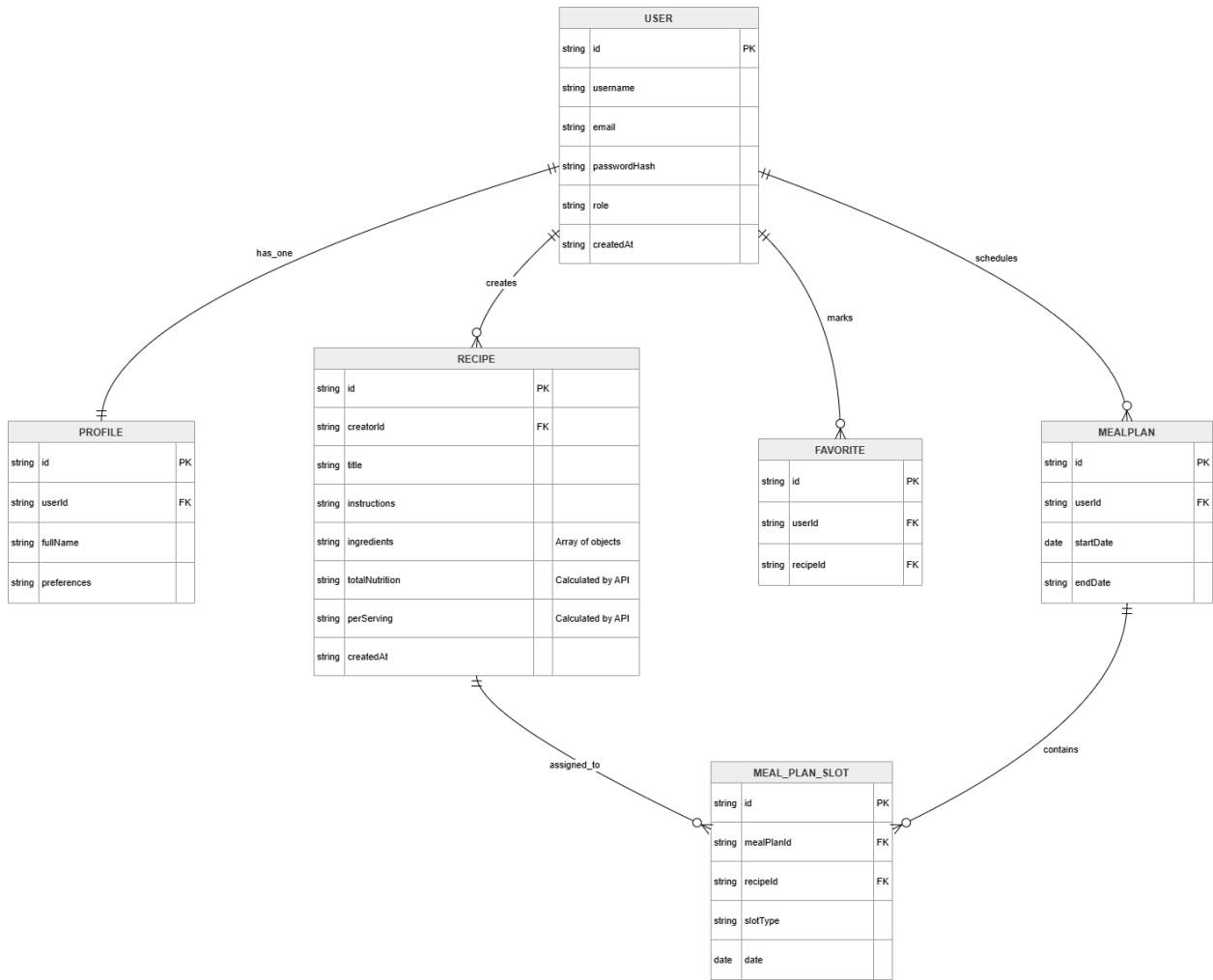
3. Frontend Dependencies

Package	Version	Purpose
react	^19.2.5	UI framework — component-based SPA rendering
react-dom	^19.2.5	DOM rendering and reconciliation
react-router-dom	^7.14.2	Client-side routing with BrowserRouter
axios	^1.15.2	Promise-based HTTP client for all API calls

formik	^2.4.9	Form state management — handles values, errors, submission
yup	^1.7.1	Schema-based form validation (used with Formik)

Package	Version	Purpose
tailwindcss	(via config)	Utility-first CSS framework; responsive design
@mui/material	^9.0.0	Material UI supplementary components
@emotion/react	^11.14.0	CSS-in-JS library (required by MUI)
html2pdf.js	^0.14.0	Export grocery list to downloadable PDF in browser

Architecture Diagrams



Security Considerations

Password Storage	Bcrypt hashing with salt rounds = 10; plain-text password is never persisted to database
Authentication	Stateless JWT; token expires in 7 days; validated on every protected route via authMiddleware
Authorization	authMiddleware applied to all mutation endpoints (POST, PUT, DELETE)
Ownership Enforcement	Recipe edit/delete verifies recipe.user === req.user.id before any modification
Input Size Abuse	15 MB JSON body limit enforced by Express; returns 413 with user-friendly message on violation
CORS	Enabled globally for development; should be restricted to known origins in production deployment

Testing Strategy

Test Type	Tool	Location	Coverage
Backend unit & integration	Jest 30 + Supertest 7	backend/src/__tests__/	tests, allpassing
In-memory DB isolation	mongodb-memory-server	Used in test setup	Isolates tests from real MongoDB instance
Frontend component tests	React Testing Library	frontend/src/	Available via npm test in /frontend